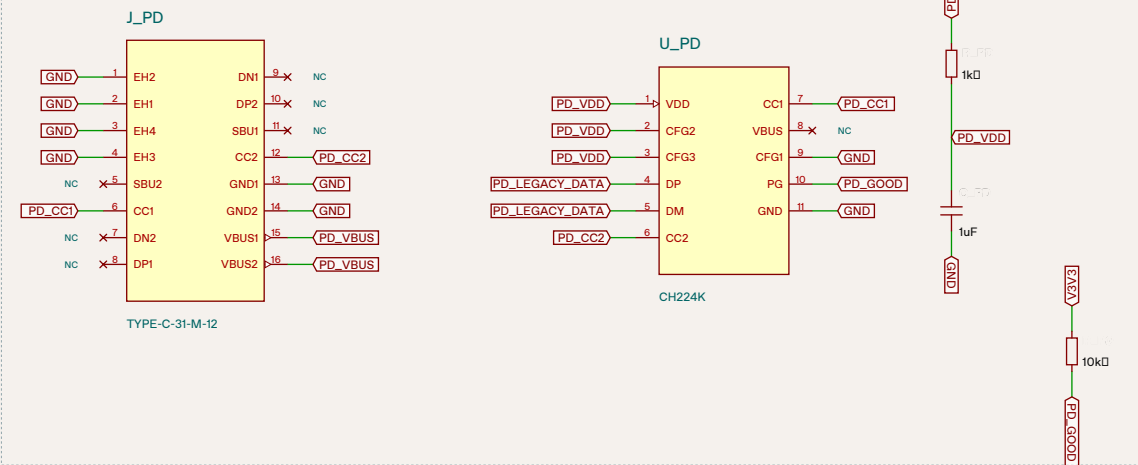


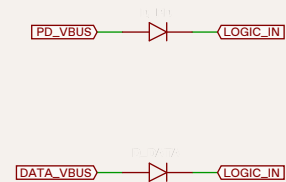
NEMA14 RP2040 | USB-C PD & Logic Power

15 V USB-C PD negotiation



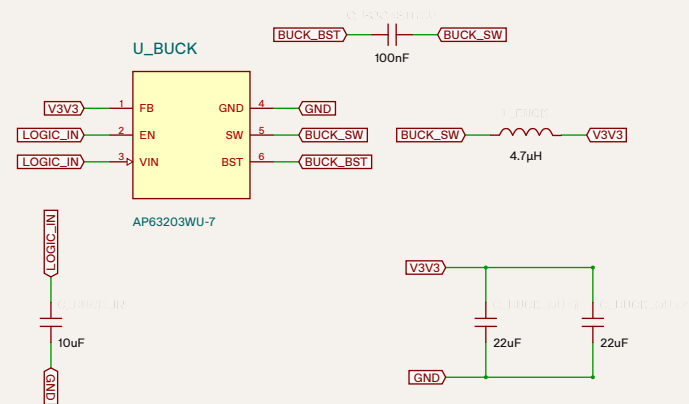
U_PD / CH224K requests 15 V (CFG1=0, CFG2=CFG3=1).
Open-drain PG is active low. DP/DM short: PD-only mode.
R_PD + C_PD supply/filter the negotiation IC.

USB / PD diode OR



Schottky diodes accept either supply and block crossfeed.

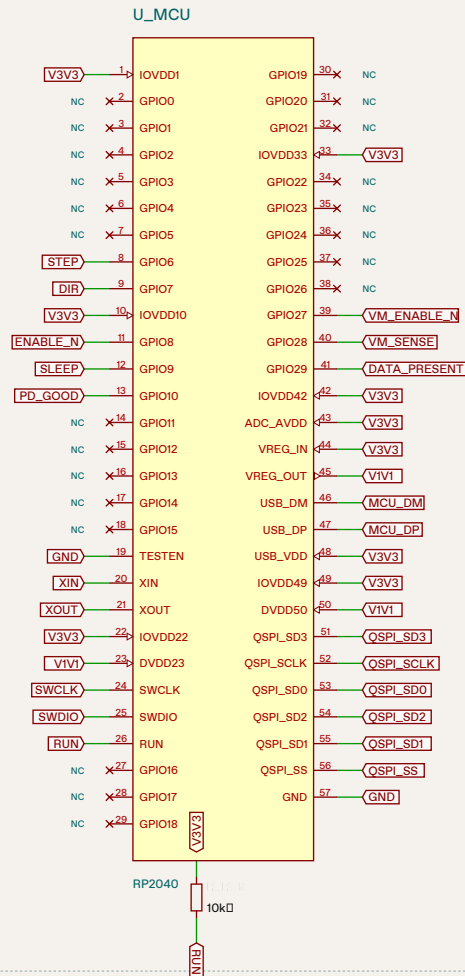
3.3 V buck supply



U_BUCK / AP63203 converts LOGIC_IN to 3.3 V.
4.7 μH L_BUCK and output capacitors filter the switching node.
C_BOOTSTRAP supports the high-side gate driver.

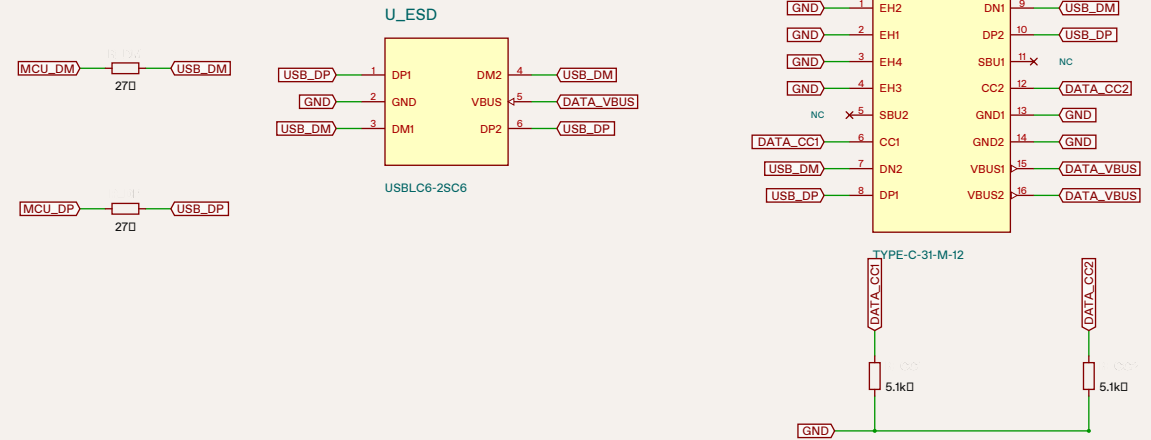
NEMA14 RP2040 | RP2040 & USB Control

Application MCU / physical pin map



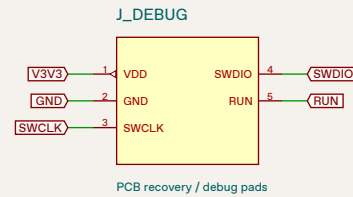
U_MCU / RP2040 generates STEP/DIR and reads power status.
Internal regulator: 1.1V core. Flash/clock/bypass: next sheet.

Computer USB-C & ESD protection

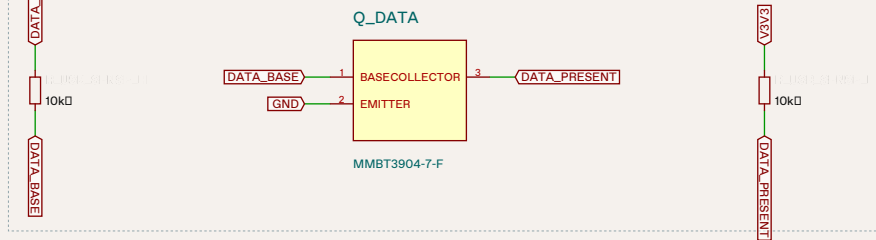


Programming & recovery

SWD / RUN pads allow debug and reset.
USB ROM boot: BOOT pads on next sheet.



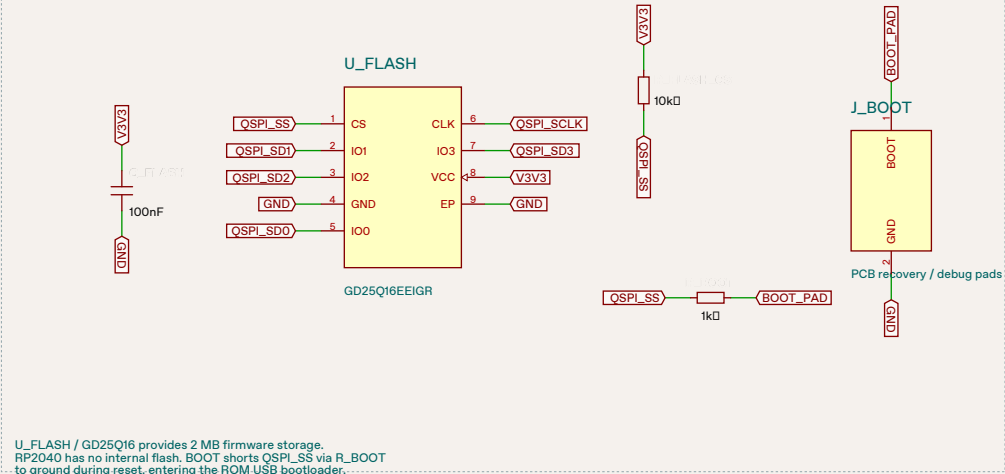
Host VBUS detection



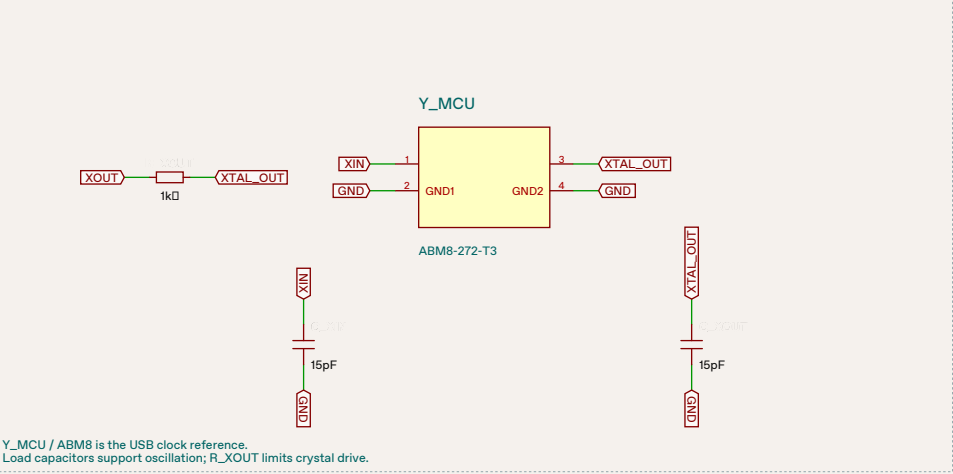
Q_DATA / MMBT3904 senses host 5V without a direct GPIO feed.
DATA_PRESENT LOW means host attached.

NEMA14 RP2040 | Flash, Clock & MCU Supply

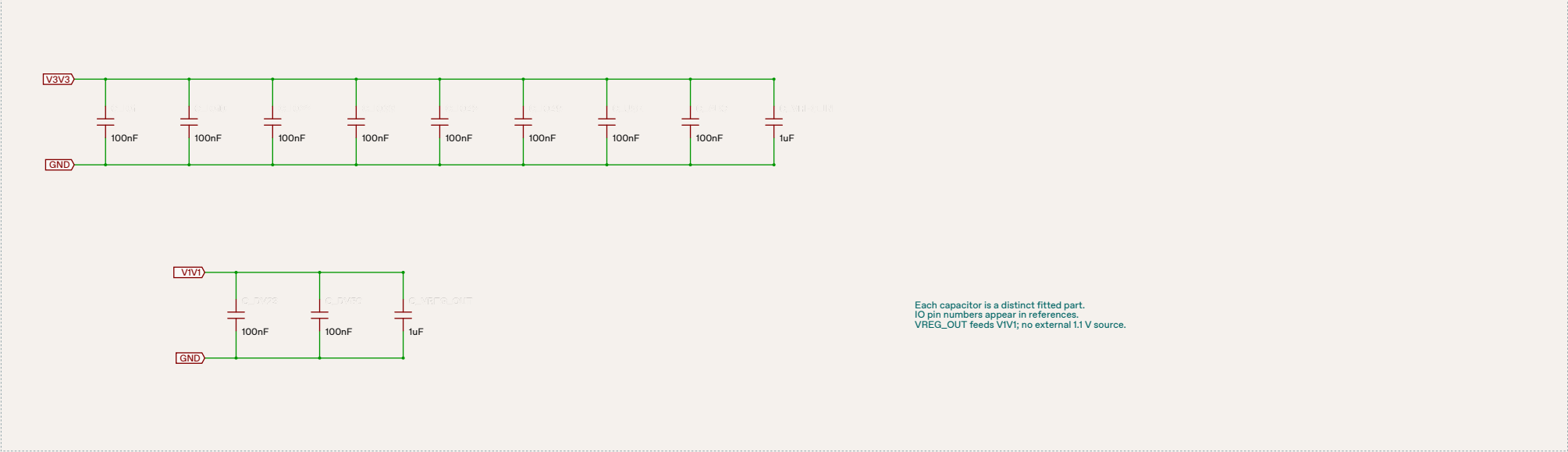
External QSPI firmware & ROM recovery



12 MHz USB clock reference

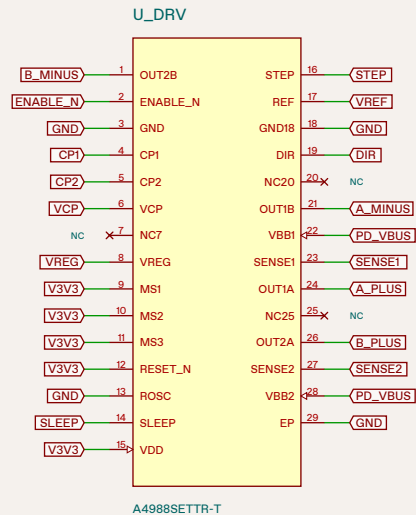


MCU power pins / local bypass

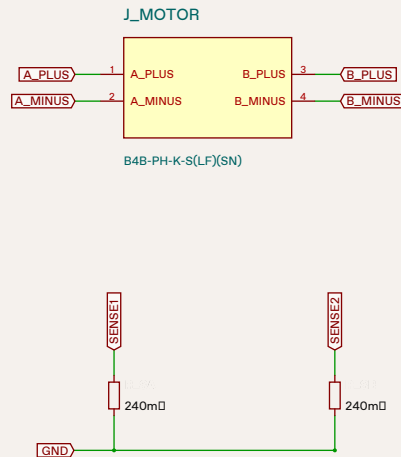


NEMA14 RP2040 | Motor Driver & Voltage Monitor

A4988 / bipolar winding driver

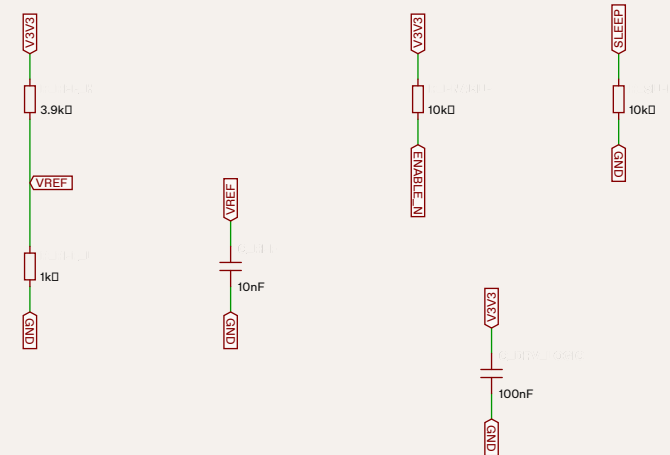


Keyed motor output & current shunts



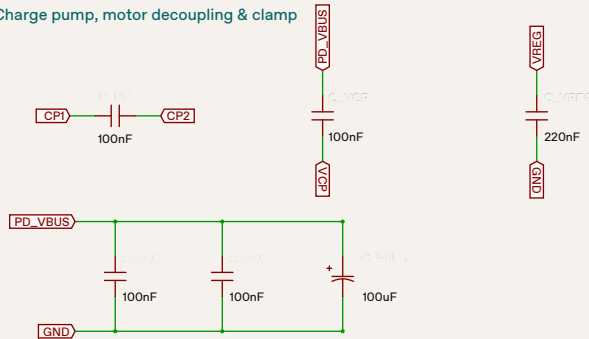
1 Black A+ 2 Green A-
3 Red B+ 4 Blue B-

Current reference & safe defaults



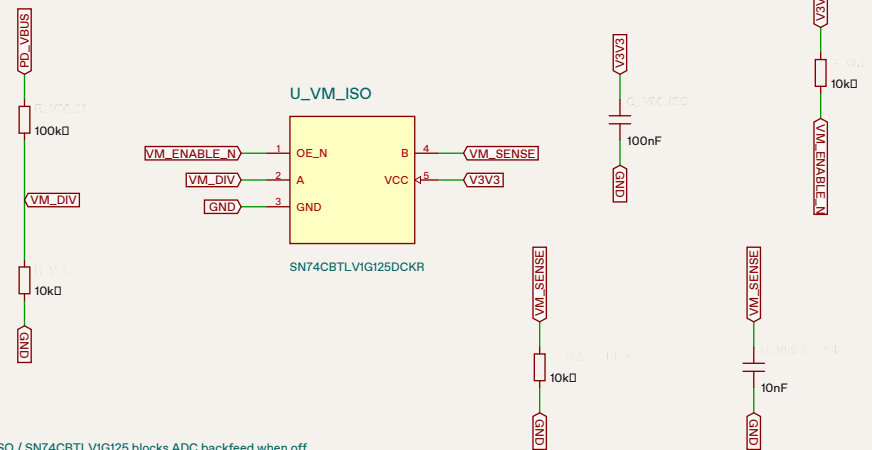
3.9k / 1k VREF and 0.24 ohm shunts: 0.351 A nominal.
ENABLE pull-up / SLEEP pull-down keep the driver off.

Charge pump, motor decoupling & clamp



U_DRV regulates both windings at fixed 1/16 microsteps.
C_CP / C_VCP / C_VREG support the internal driver rails.
Bulk capacitance and TVS limit transients; regeneration needs bench tests.

Motor-voltage monitor / power-off isolation



U_VM_ISO / SN74CBTLV1G125 blocks ADC backfeed when off.
OE pull-up defaults off; MCU enables, waits 0.5 ms, then samples VM/21.